

## Problem-Solution Fit

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|----------------------|--------------------------------------------|
| <b>Date</b>          | 01 July 2026                               |
| <b>Team ID</b>       | SWTID-2026-2030                            |
| <b>Project Name</b>  | Voice Based Concept Understanding Analyser |
| <b>Maximum Marks</b> | 5 Marks                                    |

### Problem-Solution Fit Canvas:

Based Concept Understanding Analyzer, the canvas focuses on students who need an effective way to improve conceptual understanding and communication skills. It also examines existing solutions, identifies their limitations, and presents an AI-powered solution that records voice explanations, evaluates conceptual understanding, and provides instant feedback, performance scores, and personalized learning reports.

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| <div>1. CUSTOMER SEGMENT(S) [CS]</div> <div><ul style="list-style-type: none"><li>UG/PG Students (all streams)</li><li>Students preparing for exams, viva, interviews &amp; presentations</li><li>Faculty members / teachers</li><li>Educational institutions</li><li>Online learners &amp; self-improvement seekers</li></ul></div>                                                                                    | <div>6. CUSTOMER LIMITATIONS</div> <div>EG. BUDGET, DEVICES [CL]</div> <div><ul style="list-style-type: none"><li>Limited access to quality microphone or recording devices</li><li>Unstable internet connectivity</li><li>Variations in accents &amp; pronunciation</li><li>Limited awareness of AI learning tools</li><li>Time constraints during assessments</li></ul></div>                                                                                                                                                                                                              | <div>5. AVAILABLE SOLUTIONS</div> <div>PROS &amp; CONS [AS]</div> <div><ul style="list-style-type: none"><li>Manual evaluation by teachers (Pro: Human judgement / Con: Time-consuming)</li><li>Online quizzes &amp; MCQ platforms (Pro: Easy access / Con: No speech analysis)</li><li>Speech practice apps (Pro: Practice support / Con: No concept evaluation)</li><li>Recorded lectures &amp; notes (Pro: Helpful for study / Con: No personalized feedback)</li></ul></div>     |
| <div>2. PROBLEMS / PAINS + ITS FREQUENCY [PR]</div> <div><ul style="list-style-type: none"><li>Difficulty in explaining concepts clearly</li><li>No instant &amp; personalized feedback</li><li>Manual evaluation is slow &amp; inconsistent</li><li>Lack of objective concept understanding analysis</li><li>No analysis of communication skills</li><li>Frequent before exams, vivas &amp; interviews</li></ul></div> | <div>9. PROBLEM ROOT / CAUSE [RC]</div> <div><ul style="list-style-type: none"><li>Absence of automated system to evaluate spoken explanations</li><li>Dependence on manual &amp; subjective evaluation</li><li>Lack of AI-based semantic understanding</li><li>No tool to measure concept clarity along with communication skills</li><li>Limited practice opportunities with feedback</li></ul></div>                                                                                                                                                                                      | <div>7. BEHAVIOR + ITS INTENSITY [BE]</div> <div><ul style="list-style-type: none"><li>Regularly study &amp; revise concepts (High)</li><li>Practice speaking before exams/interviews (High)</li><li>Seek feedback to improve (High)</li><li>Use online learning resources frequently (High)</li><li>Prefer tools that save time &amp; give quick results (High)</li></ul></div>                                                                                                     |
| <div>3. TRIGGERS TO ACT [TR]</div> <div><ul style="list-style-type: none"><li>Upcoming exams / viva / interviews</li><li>Project presentations &amp; seminars</li><li>Desire to improve communication skills</li><li>Need for self-assessment &amp; improvement</li><li>Competitive academic environment</li></ul></div>                                                                                                | <div>10. YOUR SOLUTION [SL]</div> <div>Voice-Based Concept Understanding Analyzer (VBCUA)</div> <div><ul style="list-style-type: none"><li>Records voice explanations (.wav via FFmpeg)</li><li>Converts speech to text using Whisper</li><li>Extracts audio features using Librosa</li><li>Evaluates concept understanding using Sentence-BERT (SBERT)</li><li>Generates scores, feedback &amp; suggestions</li><li>Creates visualizations &amp; downloadable PDF reports (ReportLab)</li><li>Helps students improve concept clarity, communication skills &amp; confidence</li></ul></div> | <div>8. CHANNELS OF BEHAVIOR [CH]</div> <div>ONLINE</div> <div><ul style="list-style-type: none"><li>Web Application (Streamlit)</li><li>Learning Management Systems</li><li>Educational Websites &amp; Platforms</li><li>Social Media / Communities</li></ul></div> <div>OFFLINE</div> <div><ul style="list-style-type: none"><li>Classroom Learning</li><li>Viva Voce / Oral Examinations</li><li>Seminars &amp; Workshops</li><li>Study Groups / Peer Discussions</li></ul></div> |
| <div>4. EMOTIONS BEFORE / AFTER [EM]</div> <div><div>BEFORE</div><div><ul style="list-style-type: none"><li>Anxious</li><li>Confused</li><li>Unsure about concepts</li><li>Low confidence</li></ul></div><div>AFTER</div><div><ul style="list-style-type: none"><li>Confident</li><li>Motivated</li><li>Clear understanding</li><li>Satisfied with feedback</li></ul></div></div>                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |